

MERIDIAN & ASSOCIATES LLP

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Case Study: Enterprise Cloud Migration

State Government Agency

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1. Engagement Overview

Client: Large US state government agency (name withheld per contract)

Industry: Public Sector - State Government

Engagement Type: Enterprise Cloud Migration & Modernization

Duration: 30 months (January 2023 - June 2025)

Team Size: 90 professionals at peak staffing

Total Fees: \$32.6 million

Lead Partner: Dr. Priya Ramanathan, CISSP, TOGAF

Meridian Offices: Washington, D.C. (lead), San Francisco, Hyderabad (GDC)

2. Client Situation & Challenge

The client, one of the largest US state government agencies by IT budget, supported over 45,000 state employees and delivered critical services to 12 million residents. The agency's IT infrastructure had accumulated significant technical debt over two decades, with the majority of systems hosted in two aging state-owned data centers. The Governor's office had mandated an enterprise cloud migration as a cornerstone of the state's Digital Government initiative, targeting measurable improvements in service delivery, security, and cost efficiency.

The agency faced a complex set of interrelated challenges:

- **Aging Infrastructure:** 60% of on-premises infrastructure was past manufacturer end-of-life, including 340 physical servers running Windows Server 2012 (end of extended support), 85 storage arrays averaging 8 years old, and network switches with known security vulnerabilities. The agency experienced an average of 14 unplanned outages per quarter, directly impacting citizen services.
- **Security & Compliance:** The state had experienced a significant ransomware incident in 2021 affecting a neighboring agency, prompting executive-level urgency around security modernization. The target environment required FedRAMP High authorization equivalent (StateRAMP), CJIS compliance for law enforcement data, and IRS Publication 1075 compliance for tax systems.
- **Application Portfolio Complexity:** The agency operated 2,500+ applications ranging from modern web applications to 25-year-old COBOL/CICS mainframe programs. An initial assessment revealed that 35% of applications had no documented architecture, 22% had no identified technical owner, and 18% were running on unsupported operating systems or middleware.
- **Workforce Readiness:** The agency's IT workforce of 1,200 had limited cloud skills, with fewer than 40 staff holding cloud certifications. Additionally, the agency operated under state civil service rules that constrained hiring timelines and salary competitiveness, making external recruitment of cloud talent extremely challenging.
- **Budget Constraints:** The migration needed to be funded within existing IT operating budgets, requiring the program to generate sufficient cost savings in early phases to self-fund subsequent phases.

The legislature had approved a \$15M one-time modernization appropriation to seed the program.

3. Meridian's Solution

Meridian designed and executed a comprehensive enterprise cloud migration program using a modified 6R framework (Rehost, Replatform, Refactor, Rearchitect, Retire, Retain) tailored to the unique requirements of government IT:

3.1 Cloud Foundation & Landing Zone

Meridian designed and deployed a secure Azure Government landing zone aligned with the Microsoft Cloud Adoption Framework and customized for state government compliance requirements. The landing zone included:

- Hub-and-spoke network topology with centralized firewall (Azure Firewall Premium) and DDoS protection
- Identity integration with the agency's on-premises Active Directory via Azure AD Connect with staged rollover
- Azure Policy guardrails enforcing StateRAMP, CJIS, and IRS 1075 controls at the subscription level
- Centralized logging and SIEM integration (Microsoft Sentinel) for security monitoring
- Automated Infrastructure as Code (IaC) deployment using Terraform modules, enabling standardized, repeatable provisioning
- Cost management framework with automated budget alerts, reserved instance optimization, and chargeback reporting by agency division

3.2 Application Migration (6R Disposition)

Meridian conducted a comprehensive assessment of all 2,500 applications and developed a disposition recommendation for each based on business criticality, technical complexity, compliance requirements, and total cost of ownership. The disposition breakdown was:

- **Rehost (Lift & Shift):** 680 applications (27%) migrated to Azure IaaS with minimal modification. Targeted applications with modern-ish architecture but no immediate business case for refactoring.
- **Replatform:** 420 applications (17%) migrated with targeted optimizations, such as moving databases from self-managed SQL Server to Azure SQL Managed Instance, or shifting web workloads to Azure App Service.
- **Refactor / Rearchitect:** 340 applications (14%) redesigned as cloud-native services using Azure Kubernetes Service (AKS), Azure Functions, and managed PaaS services. Focused on high-value, citizen-facing applications.
- **Retire:** 700 applications (28%) identified for decommissioning based on redundancy analysis, zero/low usage, or availability of SaaS replacements. Retirement of these applications eliminated \$4.2M in annual licensing and maintenance costs.
- **Retain (On-Premises):** 360 applications (14%) retained on-premises due to mainframe dependencies, specialized hardware requirements, or compliance constraints. These will be addressed in a future modernization phase.

3.3 DevSecOps Pipeline

Meridian established an enterprise DevSecOps platform to standardize application deployment, security scanning, and change management across the migrated portfolio. Built on Azure DevOps with integrated security tooling, the pipeline included:

- Automated CI/CD pipelines with mandatory stages: build, SAST (SonarQube), DAST (OWASP ZAP), container scanning (Trivy), infrastructure validation (Terraform plan), and deployment
- Pre-configured pipeline templates for common application archetypes (web app, API, batch job, data pipeline)
- Integrated change management workflow aligned with the agency's ITIL-based CAB process
- Secrets management via Azure Key Vault with automated rotation
- Compliance-as-Code validation ensuring every deployment meets StateRAMP controls

3.4 Modern Workplace (Microsoft 365 & Teams)

As part of the broader modernization initiative, Meridian led the migration of 45,000 users from on-premises Exchange 2016 and legacy file shares to Microsoft 365 (Exchange Online, SharePoint Online, OneDrive, Teams). The migration was executed over 12 weekends using a wave-based approach, with zero data loss and less than 30 minutes of mail delivery delay per user. The deployment included Microsoft Teams as the unified communications platform, replacing a fragmented mix of Cisco Jabber, Skype for Business, and ad-hoc Zoom accounts.

4. Delivery Approach

The 30-month program was organized into four phases:

- **Phase 1 - Foundation (Months 1-6):** Application discovery and assessment, landing zone design and deployment, security framework, migration factory setup, workforce training program launch
- **Phase 2 - Wave 1 Migration (Months 7-14):** Migration of 600 applications (low-complexity rehost/replatform), M365 migration, DevSecOps platform deployment, first cost savings realization
- **Phase 3 - Wave 2 Migration (Months 15-24):** Migration of 840 applications (medium/high complexity), refactoring of citizen-facing applications, application retirement execution, ongoing workforce development
- **Phase 4 - Optimization (Months 25-30):** Performance tuning, cost optimization, remaining migrations, knowledge transfer, transition to managed operations, benefits realization reporting

Team composition at peak (90 professionals):

- 1 Partner, 4 Senior Managers, 8 Managers, 24 Senior Consultants, 32 Consultants, 21 Analysts
- 55 onshore (D.C., client data centers), 35 offshore (Hyderabad GDC)
- Dedicated security team of 12 (including 3 FedRAMP-experienced assessors)
- 240 agency IT staff trained and certified in Azure fundamentals (AZ-900) and Azure Administrator (AZ-104)

5. Results & Impact

Metric	Baseline	Result	Improvement
Applications Migrated	0 in cloud	1,800 migrated	72% of portfolio
IT Operating Costs	\$52M/year	\$34M/year	35% (\$18M savings)
System Availability	99.2% (14 outages/qtr)	99.95%	Near-zero outages
Security Incidents	23/year	0 during migration	Zero incidents
Mean Time to Deploy	6-8 weeks	2-4 hours	99% faster
Cloud Certifications	38 staff	278 staff	632% increase
Applications Retired	-	700 decommissioned	\$4.2M cost avoided

The agency was recognized with the state's annual IT Modernization Award and was rated "Leader" in the National Association of State CIOs (NASCIO) State IT Modernization Index for the first time in its history. The program's self-funding model, in which early-phase cost savings funded subsequent phases, has been adopted as a reference model by three other state agencies planning similar migrations.

6. Client Testimonial

"This was the most complex technology program our agency has ever undertaken, and Meridian was the right partner at every step. They brought deep Azure Government expertise, an unwavering focus on security, and a pragmatic approach to managing our unique government constraints -- from procurement rules to workforce development. The fact that we migrated 1,800 applications with zero security incidents is a testament to their methodology and the quality of their team."

-- Chief Information Officer, Client Agency